Worksheet 6-9 C U-Substitution with Definite Integrals #1
AP Calculus Name_____

1. Evaluate: $\int x(2x+5)^8\,dx$

Directions: Evaluate the following integrals. Show all work and simplify answers as appropriate.

2. $\int_{-1}^{4} -3dx$

3. $\int_{1}^{2} \frac{2}{5-2x}\,dx$

4. $\int_{-1}^{2} 3x^2(2-x^3)^4\,dx$

5. $\int_{0}^{3} \frac{1}{(3x+1)^2}\,dx$

6. $\int_{1}^{3} e\,dx$

7. $\int_{-1}^{0} \frac{1}{\sqrt{4+3x}}\,dx$

8. $\int_{1}^{8} \frac{3}{\sqrt[3]{9-x}}\, dx$

9. $\int_{-1}^{0} x(4x^2-1)^3\, dx$

10. $\int_{2}^{4} \frac{3x}{x^2-2}\, dx$

11. $\int_{\frac{\pi}{9}}^{\frac{\pi}{3}} \cos(3x)\, dx$

12. $\int_{1}^{3} \frac{-4}{x^2}\, dx$

13. $\int_{-1}^{1} (3x)^2\, dx$

14. $\int_{0}^{2} \frac{-5x}{(x^2+1)^3}\, dx$

15. $\int_{0}^{7} \frac{3}{\sqrt[3]{x+1}}\, dx$

16. $\int_{\frac{\pi}{3}}^{\pi} -3\sin\left(\frac{x}{2}\right) dx$

17. $\int_{0}^{3} e^{\frac{x}{3}}\, dx$

18. $\int_1^4 2e^{3x}\,dx$

19. $\int_0^{e^2-1} \frac{1}{x+1}\,dx$

20. $\int_0^1 \frac{3x}{(4x^2+3)^3}\,dx$

21. $\int_{-2}^1 2x^2 e^{x^3}\,dx$

22. $\int_{\frac{\pi}{2}}^{\pi} 2\sin x\, e^{\cos x}\,dx$

23. $\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \sin^3 x \cos x\,\,dx$

24. $\int_0^1 \frac{e^x}{e^x+3}\,dx$

25. $\int_1^3 \frac{t}{\sqrt{3t^2-2}}\,dx$